Detailed Design

Member

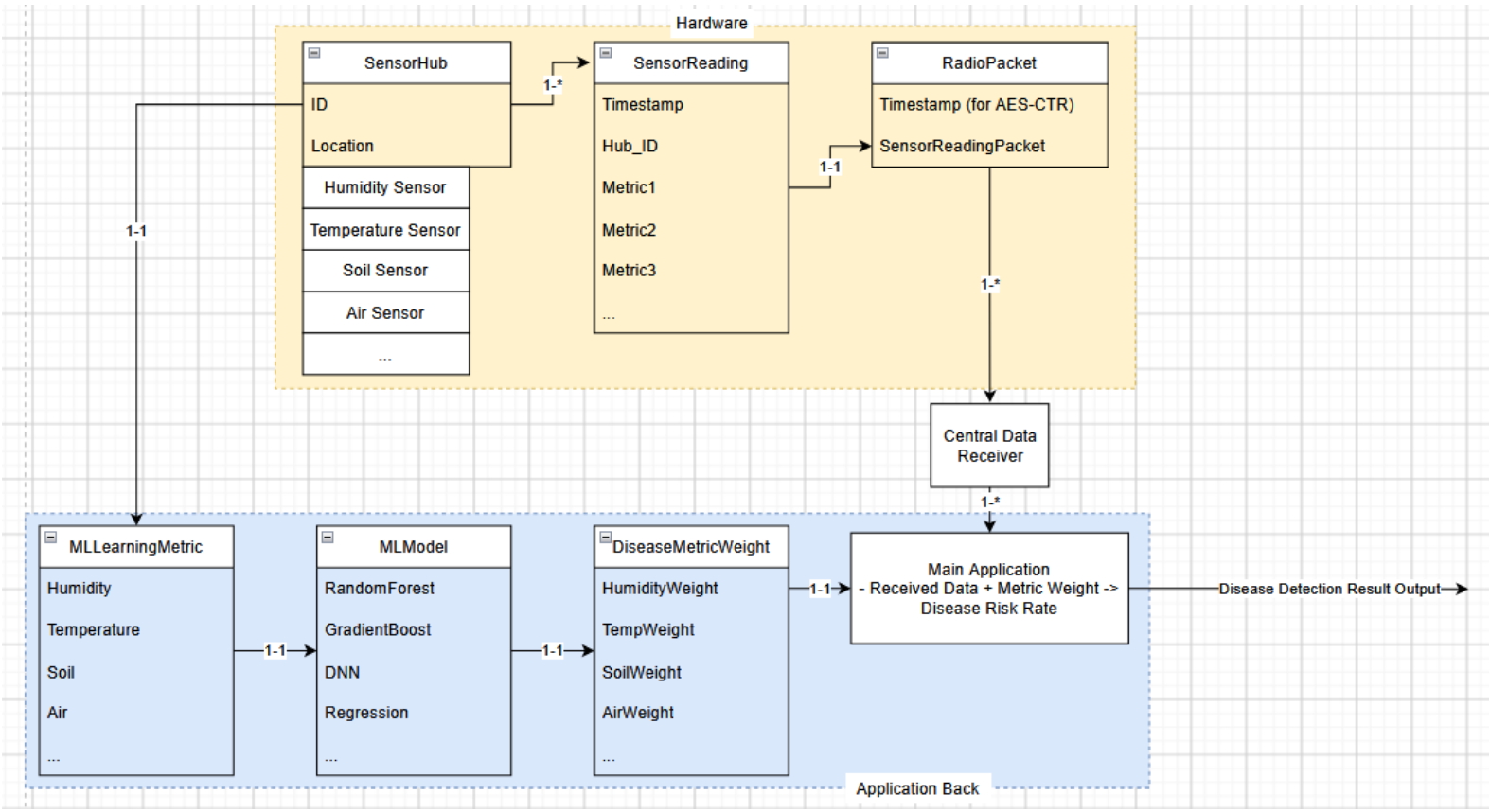
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Title, Goal, and Convention

- D1: A low-cost sensor system that include:
  - Sensor Hub: Control sensors that collect field data, read data using sensors, transmit it (using radio, or maybe another method) to a central receiver.
  - Central Application: Central Program connected to the central receiver that receive sensors data from Sensor Hub and decompress/decrypt said data. This program also hold the machine learning model and training data needed to check for disease.
  - Network: Deferred to D2
  - GUI: Deferred to D2

D1 Diagram


Hardware
SensorHub
ID
Location
Humidity Sensor
Temperature Sensor
Soil Sensor
Air Sensor
...
1-*
SensorReading
Timestamp
Hub_ID
Metric1
Metric2
Metric3
...
1-1
RadioPacket
Timestamp (for AES-CTR)
SensorReadingPacket
1-*
1-1
Central Data Receiver
1-*
MLLearningMetric
Humidity
Temperature
Soil
Air
...
1-1
MLModel
RandomForest
GradientBoost
DNN
Regression
...
1-1
DiseaseMetricWeight
HumidityWeight
TempWeight
SoilWeight
AirWeight
...
1-1
Main Application
- Received Data + Metric Weight ->
Disease Risk Rate
Disease Detection Result Output
Application Back

Structural Decisions

Decision	Choice	Reason
SensorHub	Entity	Main component that control/collect other sensor data
SensorReading	Entity	Every reading is unique in data and timestamp, and will need to be saved and used for ML
RadioPacket	Entity	Similar to above. Separated specifically for future potential choice of wireless communication model, but core aspect of packet will remain the same
MLLearningMetric, MLModel, and DiseaseMetricWeight	Entity	Each of these component of the ML model for disease detection represent a ML step associate with each other
SensorHub -> SensorReading	1 to many	A hub will create multiple reading over time
SensorReading -> RadioPacket	1 to 1	Each RadioPacket contain one SensorReading
RadioPacket -> CentralDataReceiver -> MainApp	1 to many	MainApplication will receive multiple RadioPacket containing Reading over time
CentralDataReceiver	Entity	Entity in charge of receiving data. Separated for similar reason to RadioPacket
MainApp	Entity	Entity in charge of getting Disease Detection Result from SensorReading with MetricWeight. Despite being called MainApp, this Entity act more like a bridge between these two entities

Core Algorithms

DecryptData(bytes encrypted, bytes decrypted, int size)

What it is: Function from MainApp that recrypted received DataSensor

Input: Encrypted DataSensor (bytes)

Output: Decrypted DataSensor (bytes)

Complexity: Use AES-128-CTR, O(size) with key

Reason: Simple and popular encryption choice

Edge Case: Will need to be detected outside of function, such as decrypted size too small compared to encrypted, wrong input size, wrong password, wrong nonce, etc.

Build-versus-reuse decisions

Majority of hardware (Each Sensor on Hub, Microcontroller, Wireless communication model, etc.) will be reused due to build complexity

Software (Code) will use given library from associated hardware

Components	**Choice**	**Reason**
ML Model	Reuse - Library	Too time-consuming to build one for little reason
ML Model Metric	Build	The main part of the project that need to be done manually to choose data to detect disease
Encryption	Reuse - Idea	No need for custom encryption/decryption when most publicly available choice already have highest security possible
Hardware Setup	Build	Will need to manually configurate depend on our own research environment

API contract: None

Technology Choices:

- Python was chosen for the ML part of the project due to its ease of use, and overall proficiency by members in the group (as well as being easy to get good at even without proficiency). It also have a large community library support, basically free of cost. The only trade back is a possibly lower performance compared to some other language (C++, JS, etc.), but since the computation part of the project will be done at the Central Application where power usage and speed is not as important (compared to being done at the Sensor Hub), we have decided that it is a winning trade.